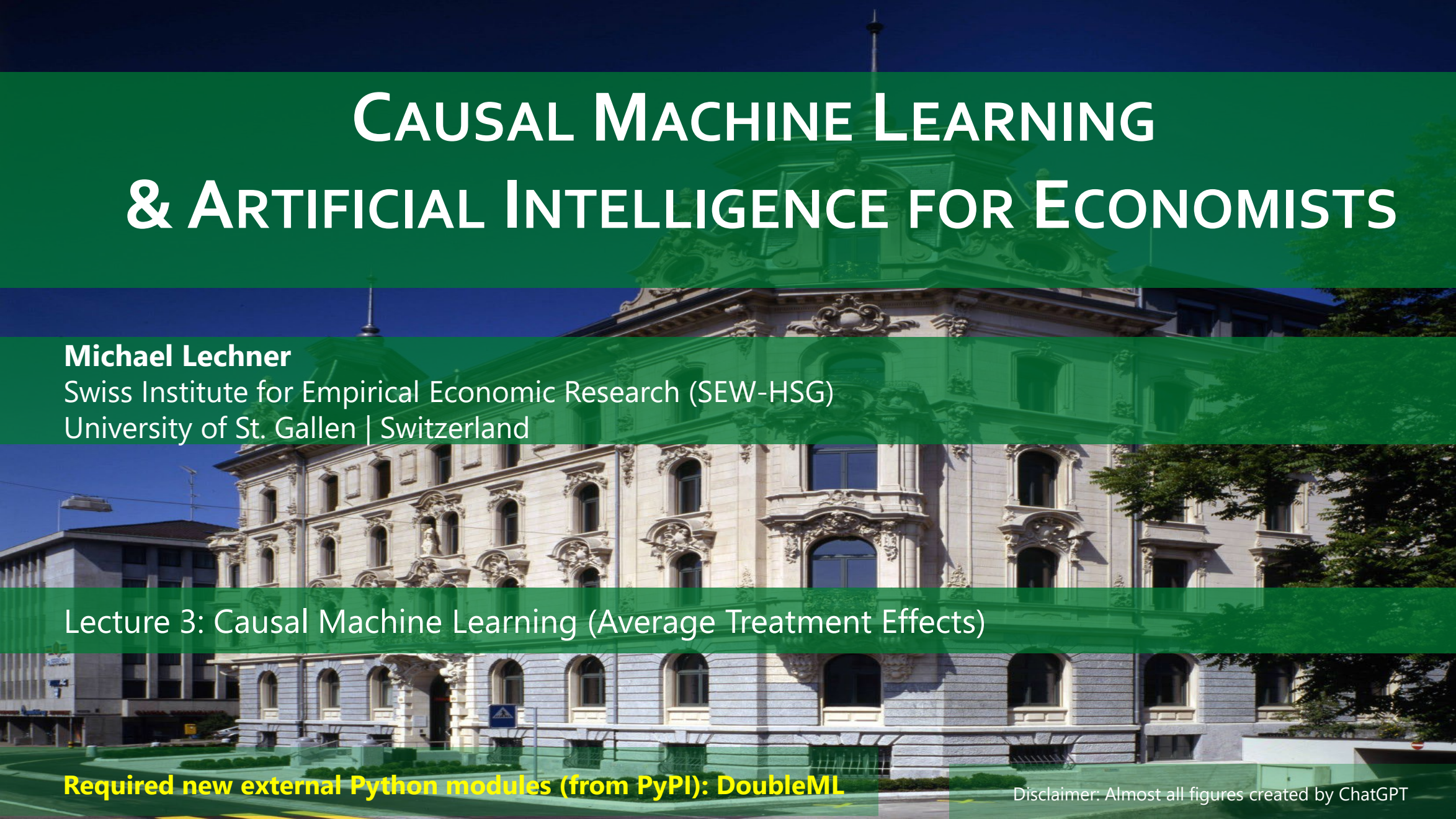


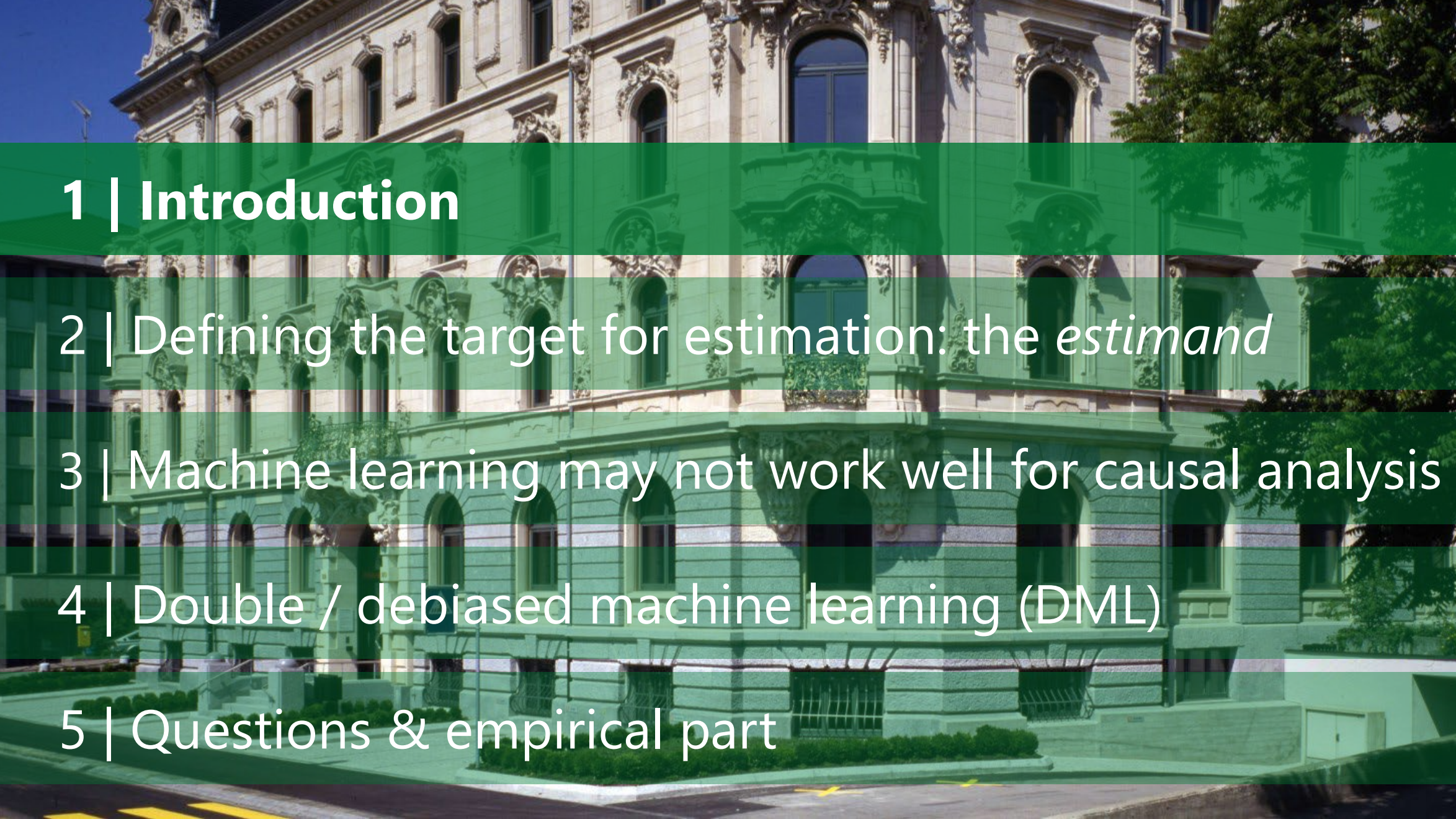
Causal Machine Learning & Artificial Intelligence for Economists

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Lecture 3: Causal Machine Learning (Average Treatment Effects)

Required new external Python modules (from PyPI): DoubleML

Disclaimer: Almost all figures created by ChatGPT



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From prediction to (average) causal effects

Lecture 2: ML methods for prediction, estimated in training data & evaluated out-of-sample

Lecture 3: ML based predictions become inputs into causal estimators

Final object is not a prediction of *Y*, but a low-dimensional causal parameter

- This section: **A**verage **T**reatment **E**ffect (*ATE*)
- Next section: **C**onditional **A**verage **T**reatment **E**ffect (*CATE(x)*)



Prediction question vs. causal question

Prediction

Who is likely to be re-employed?

Estimate $E[Y \mid X]$

Good out-of-sample prediction is the final goal

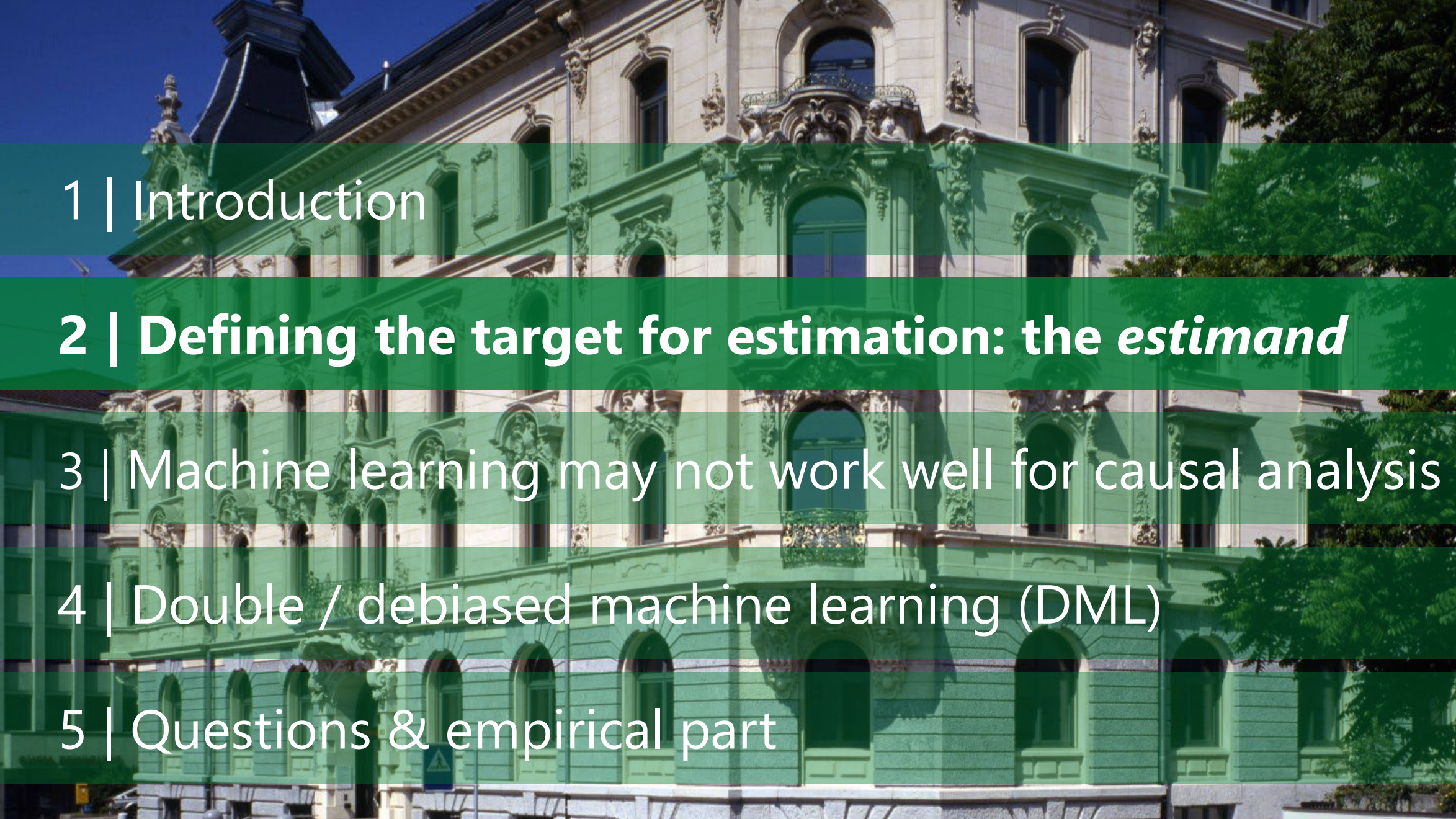
Causal effect

Does the programme increase re-employment?

Estimate $E[Y(1) - Y(0)]$

Prediction is useful only as an input into the estimation of causal parameter

Good prediction is necessary in many CML estimators, but it is not sufficient for causal inference



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Lecture 1

Data, potential outcomes & ATE (revisited)

Observed data: $w_i = (y_i, d_i, x_i)$, with $d_i \in \{0,1\}$, $i=1,\ldots, N$

- Conceptually, w_i are i.i.d. draws from a vector of random variables $W=(Y,D,X)$

Potential outcomes: $Y(1)$, $Y(0)$

Observed outcome: $Y = D\,Y(1) + (1-D)\,Y(0)$

Parameter of interest: $$ATE = \theta_0 = E[Y(1) - Y(0)]$$



Unconfoundedness | 1

Assume data is *as good as* coming from many little experiments given all possible values of *X* (stratified experiments) ➔ *Y(1)*, *Y(0)* are independent of *D for each value x of X*

Assume that *X* is not a perfect predictor of *D* (*common support)*: $0 < P(D = 1 \mid X = x) < 1, \forall x \in \mathrm{X}$

Identification idea

- Take observations of *Y* of the non-treated *with* x_i=*x* to approximate what would have happened to the treated *with value x* if not treated
- Take observations of *Y* of the treated *with value* x_i=*x* to approximate what would have happened to the non-treated *with value x* if treated

$$ATE = E_X E(Y \mid D = 1, X = x) - E_X E(Y \mid D = 0, X = x)$$



Unconfoundedness | 2

Unconfoundedness allows treated & non-treated to have different distributions of *X*

- We will remove these differences by statistical methods (e.g., Causal Machine Learning)

Unconfoundedness only holds if

- All confounders are observed (& included in X) (**C**onditional **I**ndependence **A**ssumption)
  - non-statistical knowledge needed to understand which variables are confounders
- *X* is *not* a perfect predictor of treatment (**C**ommon **S**upport)
- The measurement of *X* is not affected by the treatment (**E**xogeneity)



The estimand for the ATE | 1 | Outcome regressions

The estimand is the causal parameter expressed in terms of variables observed in the data

$$ATE = \theta_0 = E_X E(Y \mid D = 1, X = x) - E_X E(Y \mid D = 0, X = x)$$

$$\mu_0(d, x) = E(Y \mid D = d, X = x) \quad \Rightarrow \quad \theta_0 = E[\mu_0(1, x) - \mu_0(0, x)]$$

Interpretation:
Predict both expected conditional potential outcomes for everyone & average the predicted differences



The estimand for the ATE | 2 | Outcome regressions ***plus corrections***

The outcome-regression contrast is natural but fragile when the first-stage regressions are estimated flexibly

At the population level, the same ATE can also be represented by the doubly robust score

The ***doubly robust score*** adds residual correction terms using the *propensity score*,

$p_0(x) = P(D = 1 \mid X = x)$

$$\theta_0 = E[\mu_0(1,x) - \mu_0(0,x) + \frac{D(Y - \mu_0(1,x))}{p_0(x)} - \frac{(1-D)(Y - \mu_0(0,x))}{1 - p_0(x)}]$$



The estimand for the ATE | 3 | Outcome regressions ***plus corrections***

Double robustness: the score still identifies the ATE if either the outcome regression or the propensity-score model is correctly specified

Estimand based on the doubly robust score

$$\theta_0 = E[\mu_0(1,x) - \mu_0(0,x) + \frac{D(Y-\mu_0(1,x))}{p_0(x)} - \frac{(1-D)(Y-\mu_0(0,x))}{1-p_0(x)}]$$

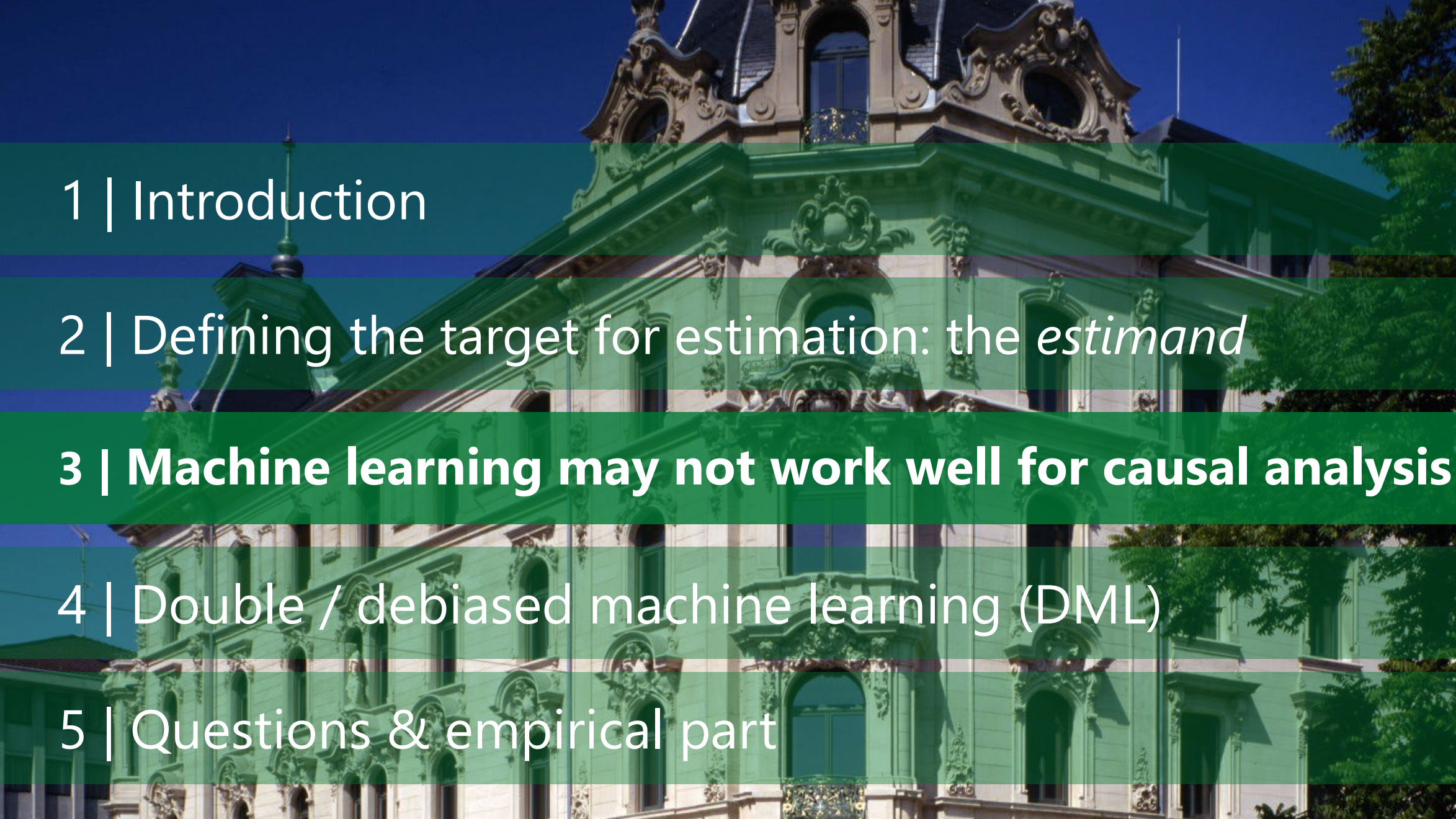
Note that the correction term does not matter in expectation (but it matters in estimation)

$$If\ \ \mu = \mu_0 : E\left[\frac{D(Y-\mu_0(1,x))}{p_0(x)} - \frac{(1-D)(Y-\mu_0(0,x))}{1-p_0(x)}\right] = 0,$$

$$If\ \ \mu = \hat{\mu}_N : E\left[\frac{D(Y-\hat{\mu}_N(1,x))}{\hat{p}_N(x)} - \frac{(1-D)(Y-\hat{\mu}_N(0,x))}{1-\hat{p}_N(x)}\right] \neq 0$$

$$E[D(Y-\mu(1,x)) \mid X=x] = p_0(x)E[Y-\mu_0(1,x) \mid D=1, X=x] = p_0(x)E[\mu_0(1,x)-\mu(1,x) \mid D=1, X=x]$$

$$E[(1-D)(Y-\mu(0,x)) \mid X=x] = [1-p_0(x)]E[Y-\mu(0,x) \mid D=0, X=x] = [1-p_0(x)]E[\mu_0(0,x)-\mu(0,x) \mid D=0, X=x]$$



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Naïve CML estimator

Replace the unknown population outcome regressions by ML estimates

Estimate $E(Y|X=x, D=1)$ with any good ML method using treated observations

Estimate $E(Y|X=x, D=0)$ with any good ML method using control observations

Predict both expected conditional potential outcomes for everyone

Average the predicted individual-level differences

$$\hat{\theta}_N^{ML-OR} = \frac{1}{N}\sum_{i=1}^{N}(\hat{\mu}_N(1,x_i) - \hat{\mu}_N(0,x_i))$$



Why prediction quality is not enough

Most ML methods are optimised for prediction loss, for example the MSE

The causal target is a low-dimensional functional of two outcome regressions

Prediction & causal analysis have different objectives

- Prediction: minimise prediction loss
- Causal analysis: minimise bias/error in the causal parameter of interest

Small systematic errors in $\hat{\mu}_N(1,x)$ & $\hat{\mu}_N(0,x)$ may barely matter for the MSE of predictions

But the same errors may matter a lot for their difference



Problem 1: Regularisation bias

Flexible ML often deliberately introduces bias to reduce variance

- Ridge shrinks coefficients
- LASSO shrinks and selects variables
- Trees and forests smooth by averaging within leaves
- Neural networks are regularised to avoid overfitting

For prediction, this bias can be useful (to minimise MSE, if inference is irrelevant)

For causal effects, it can severely bias the causal estimator & invalidate inference

Regularisation bias may be harmless for prediction but harmful for causal parameters



Problem 2: Weak overlap & extrapolation

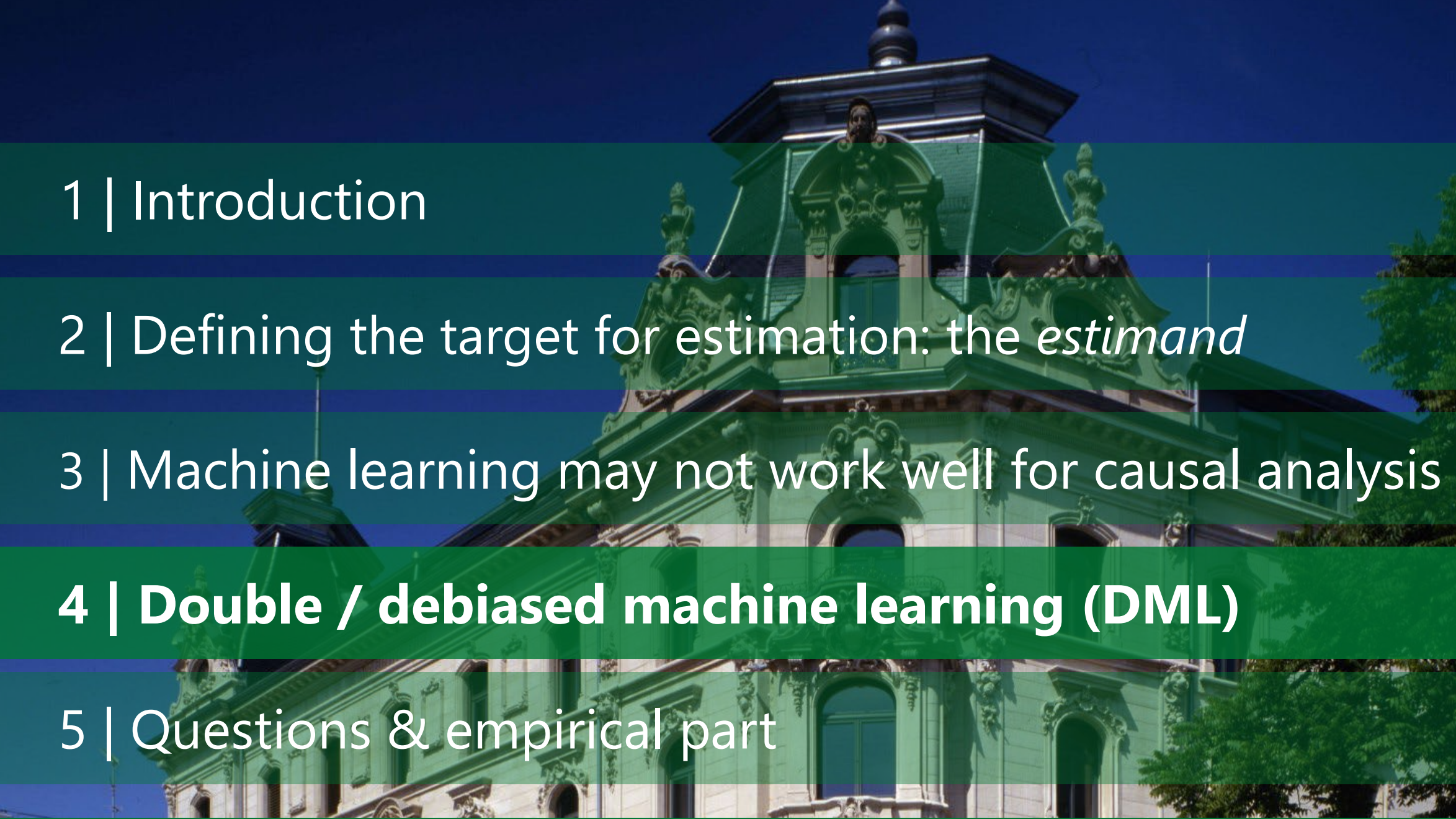
Regression adjustment predicts both potential outcomes for everybody

If $p_0(x)$ is close to 0, treated outcomes are scarce near x

If $p_0(x)$ is close to 1, control outcomes are scarce near x

Then counterfactual predictions may rely on extrapolation rather than on comparisons of observations similar w.r.t. x_i

**Weak overlap amplifies extrapolation & finite-sample instability;
lack of overlap creates an identification problem**



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How can we use flexible ML for Causal estimation without importing its regularisation bias into θ_0?

The Econometrics Journal

Econometrics Journal (2018), volume **21**, pp. C1–C68.
doi: 10.1111/ectj.12097

Double/debiased machine learning for treatment and structural parameters

VICTOR CHERNOZHUKOV[†], DENIS CHETVERIKOV[‡], MERT DEMIRER[†], ESTHER DUFLO[†], CHRISTIAN HANSEN[§], WHITNEY NEWEY[†] AND JAMES ROBINS[‖]



Double / debiased machine learning (DML) for treatment effects | 1

Main idea

- Use specific moment condition that fulfils **N**eyman **O**rthogonality **C**ondition (NOC)
  - **Here**: Dependence of moment condition on propensity score *&* outcome equations is such that small estimation errors in the nuisance functions have no relevant effect on the estimation of the causal parameter
- 1st step: Use ML to estimate *nuisance* functions
- 2nd step: Solve moment conditions given estimated nuisance functions to get structural parameter

ATE

- **Doubly Robust Score fulfils the NOC**

Structural parameter: Low dimensional object of interest

Nuisance parameters: Potentially high-dimensional object helping to estimate structural parameters – not interesting in itself



DML for treatment effects | 2 | Basic idea

Step1: Estimate unknown functions $\mu_0(.)$ **&** $p_0(.)$ by standard predictive ML

Step 2: Combine these estimates so that the estimator for ATE has good properties

- Based on moment conditions that fulfil the *Neyman Orthogonality Condition*
  - $W=(Y,D,X)$;
  - Vector η_0 contains the *nuisance* parameters $(\mu_0(.), p_0(.))$
  - $\psi(W, \theta, \eta)$ is function of data & parameters
  - Moment condition: $E\psi(W,\theta_0,\eta_0)=0$
  - Neyman orthogonality condition (NOC): $\partial_\eta E\psi(W,\theta_0,\eta)\,|_{\eta=\eta_0}=0$



DML for treatment effects | 3

NOC-compatible moment condition for the ATE – doubly robust score

$$\psi(W,\theta,g,m) = \mu(1,X) - \mu(0,X) + \frac{\big(Y-\mu(1,X)\big)D}{p(X)} - \frac{\big(Y-\mu(0,X)\big)(1-D)}{1-p(X)} - \theta$$

- NOC implies that *small* estimation errors for $\mu(d,x)$ & $p(x)$ are less relevant



DML for treatment effects | 4

DML estimator for the ATE

$$\hat{\theta}_N^{DML} = \frac{1}{N}\sum_{i=1}^{N} \hat{\mu}_{-i}(1,x_i) - \hat{\mu}_{-i}(0,x_i) + d_i \frac{y_i - \hat{\mu}_{-i}(1,x_i)}{\hat{p}_{-i}(x_i)} - (1-d_i)\frac{y_i - \hat{\mu}_{-i}(0,x_i)}{1-\hat{p}_{-i}(x_i)}$$

- The nuisance functions $\hat{\mu}_{-1}(1,x_i), \hat{\mu}_{-1}(0,x_i), \hat{p}_{-1}(x_i)$ are estimated without using observation '*i*'
  - Avoids own-observation overfitting & greatly simplifies the asymptotic analysis

Under regularity conditions, this estimator is √N-consistent, asymptotically normally distributed & semiparametrically efficient

$$\sqrt{N}\frac{\hat{\theta}_N - \theta_0}{\sqrt{E\left[\psi(Y,X,D,\theta_0,g_0,m_0)^2\right]}} \xrightarrow{d} N(0,1)$$



DML for treatment effects | 5 | Implementational issue

How to implement "*without using observation i*" ?

CCDDHNR (2018) suggest implementing this estimator by k-fold sample splitting ("cross-fitting")

- Repeat for every fold *k=1:K*
  - Estimate nuisance functions on the remaining *K-1* folds (without using data from fold *k*)
  - Estimate *ATE* for fold *k* (by evaluating the estimated nuisance function with the data, *x*, of fold *k*)
- *Final estimator is computed as average of these K estimators*
  - Estimator is consistent, efficient, √*N*-convergent & asymptotically normal
- Approximate standard error of √*N* x estimator is computed as $\hat{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N}\hat{\psi}_i^2, \quad st.err.(\hat{\theta}_N^{DML}) \simeq \frac{\hat{\sigma}^2}{\sqrt{N}}$



DML for treatment effects | 6

Pros

- Theoretical properties are known & attractive
- Only the product of the estimation errors of the outcome regressions & the propensity scores must vanish faster than $N^{-1/2}$
  - A useful benchmark: if both converge faster than $N^{-1/4}$, this condition is satisfied. A slower rate for one nuisance estimator can be compensated by a faster rate for the other one
- Allows choosing ML estimator known to work 'best' for specific prediction problem

Cons

- Weighting by $1/p(x)$ and $1/(1-p(x))$ may be unstable if estimated $p(x)$ gets close to 0 or 1. This may occur if
  - ML estimators predict $p(x)$ very well such as to (almost) separate treated & controls
  - the (real) common support is weak



Double / debiased ML (DML) protects the causal parameter

Naïve ML outcome regression

Estimate $\mu_1(x), \mu_0(x)$

Plug nuisance estimates directly into a non-orthogonal plug-in estimator

Same data may be used for training & function evaluation

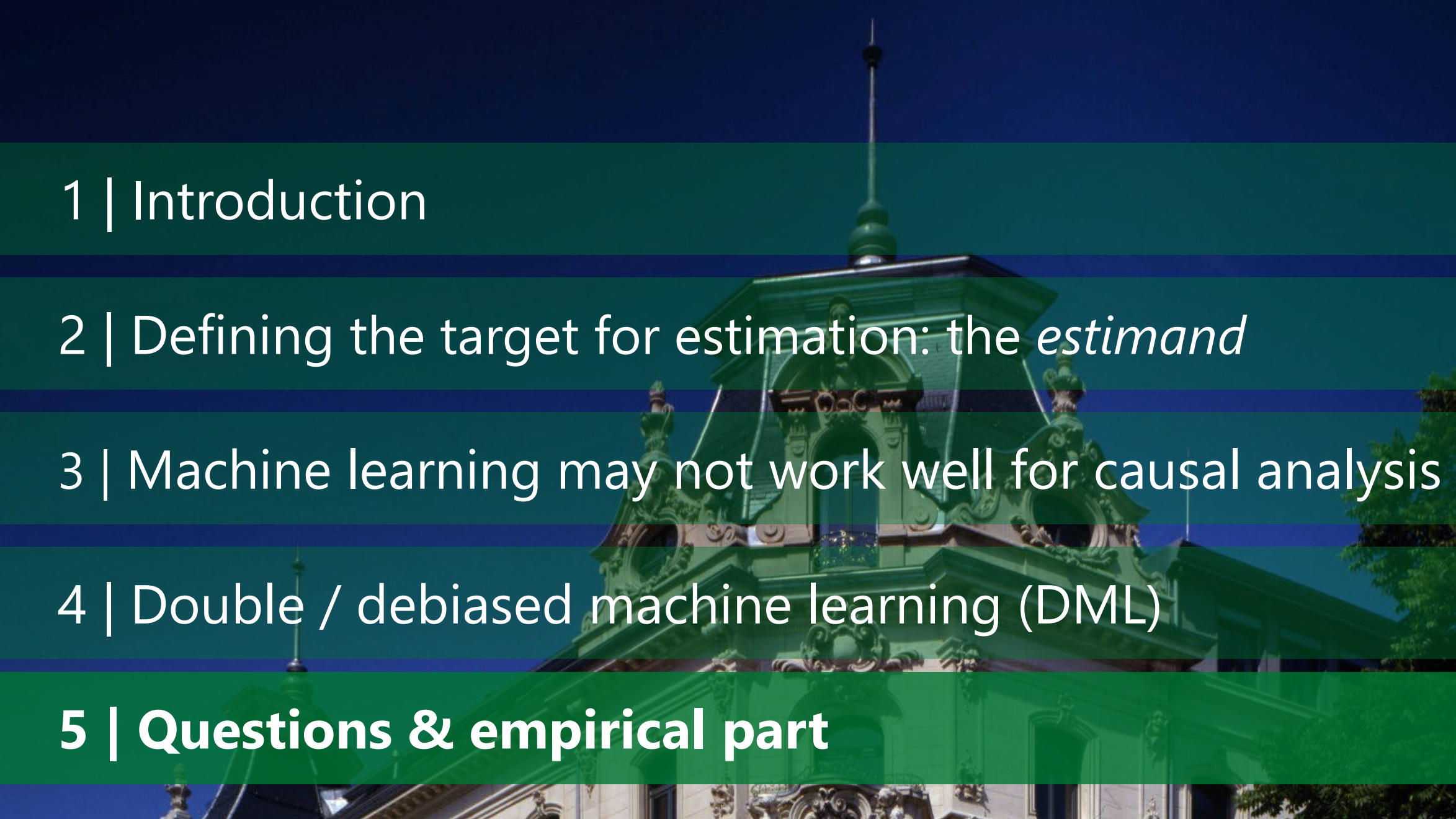
Double / Debiased Machine Learning

Estimate $\mu_1(x), \mu_0(x), p(x)$

Use an orthogonal score + cross-fitting

Cross-fitting gives out-of-sample nuisance predictions

DML is not a new machine learner; it is a disciplined way to use machine learners inside causal estimators



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Grades

Group presentation of empirical projects (50%) in last lecture

- Tasks will be distributed in beginning of last lecture
- Each group solves problems allocated to them & prepares a presentation
- Presentations will be held in this lecture as well (graded)
- **Group members are to be decided not later than after 3rd lecture (Nov, 23, 15:00)**
  - **If you do not belong to a group, you are a one-person group (designated group-leader should send an email to Federica.Mascolo@unisg.ch with names of all group members, & all group members in cc)**

Oral exam on Thursday, Dec., 17 (50%)



Rules for group work (including graded group exercise)

Groups may have up to 4 members

- *All group members must agree to all group members*

Group memberships may change until lecture 3

You are free to use any AI you like

- Graded exam: Contact with humans is prohibited & will be monitored (& will be useless )

Lecturer is available for discussions & help during group work

- limited in graded group work

You should always ***install & test*** the required packages ***before the lecture***

- ***Required (additional) packages are always shown on 1st slide of each lecture***



Empirical group work | 1

Today, we use *ALMP_data.csv* for the effect estimation of active labour market programmes. You find additional information about the programmes in the file *ALMP_inst_info.pdf*.

Although there are four different programme types, we aggregate them into one group, i.e., the goal is to estimate the effect of participating in the active labour market programme in global.

Before starting the estimation, argue why the Unconfoundedness assumptions is likely to hold. What makes this assumption plausible, & what are potential concerns?



Empirical group work | 1

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Empirical group work | 2

Estimate the ATE of participating in the ALMP using the DoubleML package. As outcomes consider earnings & being employed in the 4th quarter of 19X9

Are the effects the same for men and women?

Investigate the sensitivities of the results for the population with respect to the choice of first stage estimators (use those estimators considered the regression part of section 2)



Next lecture

Causal heterogeneity & causal forests